Pinkie Pie's Balloon Palace

The Balloon Shop inside a glass sphere: an Ambisonics encoder, 1st to 7th order.

Manual for version 0.3.1



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About this manual

Pinkie Pie's Balloon Palace is the Balloon Shop inside a glass sphere: an Ambisonics encoder, from 1st to 7th order (3rd by default). Stereo comes in; a sound field comes out (16 channels at 3rd order, 36 at 5th, 64 at 7th), for a decoder to play on loudspeakers or headphones. Little bouquets of balloons fly round inside the sphere, each carrying its own copy of the effects, and every sound is a single point you can hear and find, with its distance, its Doppler shift and one reverb for the whole sphere.

This manual covers version 0.3.1. Its middle part, the guide, is the same text you'll find in the plug-in's library: click **help** in the top right corner. Before it come the installation steps, how to route it to a decoder, a map of the window and how the sound flows. After it there are the flight modes, distance and Doppler, a gesture reference, every parameter your DAW can automate, the store's effects, the presets and answers to common problems.

Everything in the Balloon Shop is here too, so the shop's own manual also applies.

Installing

Download it from the Balloon Shop's Releases page (its releases are tagged `balloonpalace-v...`): `PinkiePiesBalloonPalace-macOS.zip` (macOS 11 or later, Apple silicon and Intel), `PinkiePiesBalloonPalace-Windows.zip` (Windows 10 or later, 64-bit) or `PinkiePiesBalloonPalace-Linux.tar.gz` (x86-64). macOS gets an Audio Unit, a VST3 and a standalone app. Windows gets a VST3 and a standalone app, and Linux adds an LV2.

macOS by hand

1. Unzip `PinkiePiesBalloonPalace-macOS.zip`.
2. Copy `Pinkie Pies Balloon Palace.component` to `~/Library/Audio/Plug-Ins/Components` and `Pinkie Pies Balloon Palace.vst3` to `~/Library/Audio/Plug-Ins/VST3`. Put the app anywhere you like.
3. The builds aren't notarised by Apple, so macOS may refuse to open them the first time. In Terminal:

   ```
   xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/Components/"Pinkie Pies Balloon Palace.component" \
     ~/Library/Audio/Plug-Ins/VST3/"Pinkie Pies Balloon Palace.vst3" "/Applications/Pinkie Pies Balloon Palace.app"
   ```

4. Rescan plug-ins in your DAW. It appears under **FreakiePie**.

Windows

Unzip, copy `Pinkie Pies Balloon Palace.vst3` to `C:\Program Files\Common Files\VST3` and rescan. The standalone `Pinkie Pies Balloon Palace.exe` runs from anywhere. SmartScreen may warn you the first time because the build isn't signed: choose *More info*, then *Run anyway*.

Linux

```
tar -xzf PinkiePiesBalloonPalace-Linux.tar.gz
mkdir -p ~/.vst3 ~/.lv2
cp -r "Pinkie Pies Balloon Palace/Pinkie Pies Balloon Palace.vst3" ~/.vst3/
cp -r "Pinkie Pies Balloon Palace/Pinkie Pies Balloon Palace.lv2" ~/.lv2/
```

Then rescan in your DAW. The standalone app runs with `"./Pinkie Pies Balloon Palace"` and uses ALSA or JACK.

Building it yourself

You need CMake 3.22 or later and a C++20 compiler (Xcode, Visual Studio 2022, or GCC or Clang).

```
git clone --recursive https://github.com/femifleming/PinkiePiesBalloonShop.git
cd PinkiePiesBalloonShop
cmake -S BalloonPalace -B build-balloonpalace -DCMAKE_BUILD_TYPE=Release
cmake --build build-balloonpalace --config Release --target BalloonPalace_All
```

Spatial audio: what comes out, and where it goes

The Balloon Palace is an encoder. It never plays straight to loudspeakers or headphones: it puts out a sound field, and a decoder after it turns that into speaker feeds or binaural headphones. That's what lets the same session play on any speaker layout.

Order	1st to 7th, set with the order token (3rd by default)
Channels	$(\text{order} + 1)^2$: 4, 9, 16, 25, 36, 49 or 64
Channel order	ACN (0 to 15)
Normalisation	SN3D, without the Condon-Shortley phase (AmbiX)
Axes	x ahead, y to the left, z up (as in the IEM plug-ins)
Input	Stereo
VST3 layout	Ambisonics (ACN) up to 7th order, so hosts that know Ambisonics tracks offer it there; the Audio Unit publishes 3rd order, and takes the others if the host asks

In REAPER with the IEM plug-ins

1. Pick the order with the order token (bottom right; 3rd by default).
2. Give the track as many channels as that order needs, with the track's **Route** button, *Track channels*: 16 for 3rd order, 36 for 5th, 64 for 7th.
3. After it, add a decoder at the same order: the IEM **AllRADecoder** for a loudspeaker layout, or the **BinauralDecoder** for headphones (SN3D). Add the IEM **EnergyVisualizer** too if you like.
4. To send several Balloon Palace tracks to one decoder, give them all the same order, feed a bus with the same number of channels, and decode once on the bus.

Any host that can run a plug-in on that many channels works the same way. SPARTA's decoders work too: choose ACN and SN3D. On a track with too few channels for the order you picked, the highest order that fits comes out, and the map says so.

Which order?

A higher order pins every sound down more sharply. Through a decoder of the same order, a single point puts about 36 % of its energy within 20 degrees of where it is at 3rd order, 62 % at 5th and 83 % at 7th. You hear that most on a big dome of loudspeakers; through headphones the difference is small. It costs a little more: 16 grains with the reverb and the weather take about 23 % of one core at 3rd order, 26 % at 5th and 31 % at 7th (an Apple M2 Max at 48 kHz). The first time you pick 4th order or above, the sound pauses for a moment while the snow and the heatwave get ready for the extra channels.

The one exception to "encoder only" is your own listening while you work: that's the decoder's job, so there's no headphone mode in the Balloon Palace itself.

The window

The numbers on the picture match the list under it. The window can be resized from its corner; everything scales together.



1. The weather row: cloudy, rain, snow, heatwave, fog, wind, hail, aurora, twister and auto weather, all the same size; one that is on is lit up. Right-click the snow or the heat for their fade times.
2. The balloon store.
3. Pop sounds, help (the library) and reset.
4. The sonic rainboom: click for one, right-click for what sets it off.
5. Gumballs: the gumball menu, every control again as a gumball that stays put.
6. The pink pump: blows up plucked notes.
7. The teal pump: catches sounds into frozen balloons. Next to it, the next note's pitch.
8. The helium tank and the canister rail along the bottom.
9. The chalkboard: grain shapes and the master EQ.
10. The pin: drag it to pop a balloon.
11. What the teal pump catches: the effects or the input.
12. Pinkie.
13. The sphere: drag the empty glass to turn it, scroll or pinch to zoom, double-click to reset the view.
14. The energy map: where the sound is loudest, from the front.
15. The jar of grains: click for another, drag one into the sphere.
16. The tokens: flight, speed, distance, deviate, grain mode, Doppler and the Ambisonic order.
17. Move the cursor here and the card comes down: presets, the scale and the bypass. At the left edge, the order list slides out.

How the sound flows

In	Stereo in: your input, the plucked notes and Trixie, as in the shop.
Your dry sound	Encoded once, as a point at the little pink loudspeaker, through the distance model.
The grains	On spatial, every grain runs its own copy of the effects and plays only what its effects add, as a point at the grain. On scatter, one copy. On copies, every grain is the whole shop. Copies are decorrelated so they never phase against each other.
The granulator	A cloud of short grains, each born at its own direction round one of the grains.
Pops and frozen balloons	From the loudspeaker's place.
Sonic rainboom	The dive and the boom as points, the rainbow as seven rings round where it broke.
Hail, rainbow, aurora	Made from everything the sphere is playing: hailstones as points bouncing down from overhead, the rainbow as seven arcs across one side of the sky, the aurora as four curtains high round you.
Sphere reverb	One 64-line FDN for the whole sphere, run on every channel. Grains further out send more into it.
Energy map	Measured here (from the first 16 channels, whatever the order), for the glass and the map at the top left.
Master	A twister, the wind and the fog on every channel alike (so nothing moves from where it is), then weather, heat, snow and the chalkboard EQ on every channel, then the leveler and the limiter (-1 dBFS), linked across the channels.
Out	Ambisonics at the order you picked (ACN, SN3D: 16 channels at 3rd order, up to 64 at 7th), to a decoder.

The guide

The chapters of the library inside the plug-in, with pictures. Each starts on its own page.

Welcome

Every balloon in the shop is a control, and how full it is shows its value.

In short: drag to move, scroll to inflate, shift-drag for fine changes, double-click to reset, click a caught sound to type its pitch, drag the pin to pop them.

Drag a balloon to move it, and let go to watch it float back up on its string. Scroll or swipe over it to blow it up or let some air out. Hold shift and drag up or down to set it by hand (hold alt as well for finer steps). Double-click to put it back how it started.

Balloons with a setting written on them, like shape or octave, step to the next one when you click. Switches like freeze turn on and off with a click.

The bows hold each bunch. Drag a bow to move its whole bunch. The next page shows which colour belongs to which effect.

The sphere

The shop is inside a glass sphere now, and you're the little head in the middle, looking ahead into the screen. The bows stand on a ring round its floor, in the order the sound goes through the effects, and their balloons float up inside it.

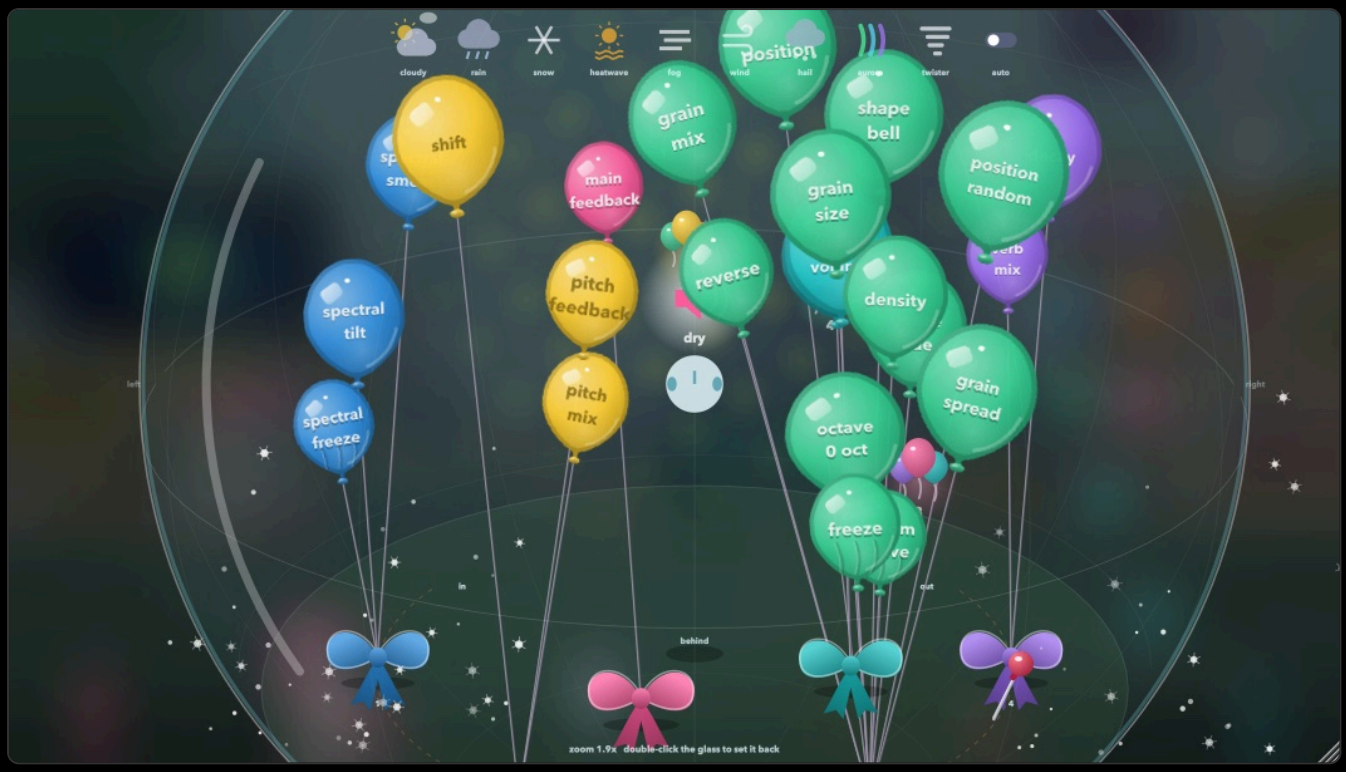
Drag the empty glass to turn the sphere round and look at it from anywhere. Turning it only changes your view, never the sound.

Scroll or pinch on the empty glass to zoom in, towards wherever the cursor is: the balloons grow so they're easy to read and edit, and the room behind blurs and darkens. Double-click the glass to set the view back.

The words on the glass show which way is ahead, left, right, behind and up. The glow on the glass and the map at the top left show where the sound is loudest, like the IEM EnergyVisualizer: red is the loudest place, then yellow, green and blue.



The sphere turned to look at it from above.



Zoomed in: the room behind blurs and darkens.

Grains in the sphere

The little bouquets flying in the sphere are grains. Click the jar on the left for another one, or drag one out of the jar and drop it where you want it. Drag a grain back onto the jar, or right-click the jar, to take one away. You can have up to 16.

Your own sound, dry, plays from one place: the little pink loudspeaker. Drag it anywhere in the sphere.

The grains token on the right picks how the effects use the grains. On spatial, every grain carries its own copy of the effects (the smear, the pitch shifter, the gumball machine and what you bought), and each copy's sound plays at its grain as a point, so you can hear where each one is. The granulator (the mint balloons) is a cloud of short grains, each one born at its own direction round one of the grains, like the IEM GranularEncoder: grain spread sets how far round its grain each one may land, from exactly there to anywhere on the sphere. On scatter there's only one copy of the effects, at the first grain, and the granulator's grains still go to all of them. On copies, every grain is the whole shop, your dry sound and all, the way the sphere first worked.

Deviate is how far the copies stray from your settings. The first grain always plays your settings as they are; the others drift away slowly, each its own way, and now and then freeze or let go by themselves. In the granulator it makes every grain's length, start and pitch a little different. Every sound in the sphere is a single point, so you can always tell where it is.

A grain glows with its sound, hops when it stutters and frosts over when it freezes.



Scatter: one copy of the effects, its granulator grains flying to every grain.



Six grains flying the Lorenz attractor.

Flying

The flight token picks how the grains move, and speed sets how fast.

Still keeps each grain where you dropped it. Orbit sends each one round you on its own tilted ring. Lorenz flies them along the Lorenz attractor, the butterfly, each one starting somewhere else on it. Flock makes them a flock of birds (boids): they keep apart, line up and stay together. Random picks a new place every so often and glides there. Spiral winds them round you, climbing and sinking. Lissajous traces looping figures, each grain at its own rates. Rossler flies them on the Rossler attractor, a loose coil that now and then flicks up. Breathe sends them out to the glass and back in again, like breathing. Fall drops them from the top of the sphere to the bottom, over and over. Ping-pong throws them straight across the sphere, from one side to the other.

Four flights follow Bernhard Leitner, who since 1968 has built rooms out of sound moving along lines of loudspeakers. Line runs each grain along its own straight line through the room and through you, from wall to wall and back at an even pace. Pendulum hangs each grain from a point overhead and swings it through underneath you, slow at the ends and quick in the middle. Column makes a vertical space: each grain climbs and sinks along its own column standing round you. Soundcube is his Soundcube of 1969, a cube with a grid of loudspeakers on every wall (nine to a wall here). The grains don't fly. They step from one loudspeaker to the next, drawing a ring round you and then another one higher or lower.

Five follow Iannis Xenakis. Sonotron is the accelerator of sound particles from Terretektorh, and the turning drums that end Persephassa: all the grains on one ring that turns faster and faster along a logarithmic spiral as it widens, then starts again the other way. Paraboloid glides every grain along its own straight line, and the lines together draw the saddle of the Philips Pavilion, as the string glissandi of Metastaseis do. Gas is Pithoprakta: the grains are the molecules of a gas, each flying straight at a speed drawn from the Maxwell-Boltzmann distribution until it hits the glass and leaves at a new one. Nomos puts the grains on the corners of a cube that turns from one of its 24 rotations to the next, each the product of the two before it, as in Nomos Alpha. Screens cuts the sphere into eight cells and moves each grain from cell to cell by a Markov chain, with the two transition tables of Analogique A.

Distance is the size of the bubble you're in, from 1 metre to 50. In a small bubble everything is close and loud and the room answers quickly; in a big one everything is far away, quieter, a little duller and roomier. It sounds the way distance does, after the IEM RoomEncoder: the sound takes longer to arrive from further away (so anything flying towards you or away bends in pitch, like a passing car), and the walls of a room round the bubble answer each sound faintly. Close in, a sound spreads round your head.

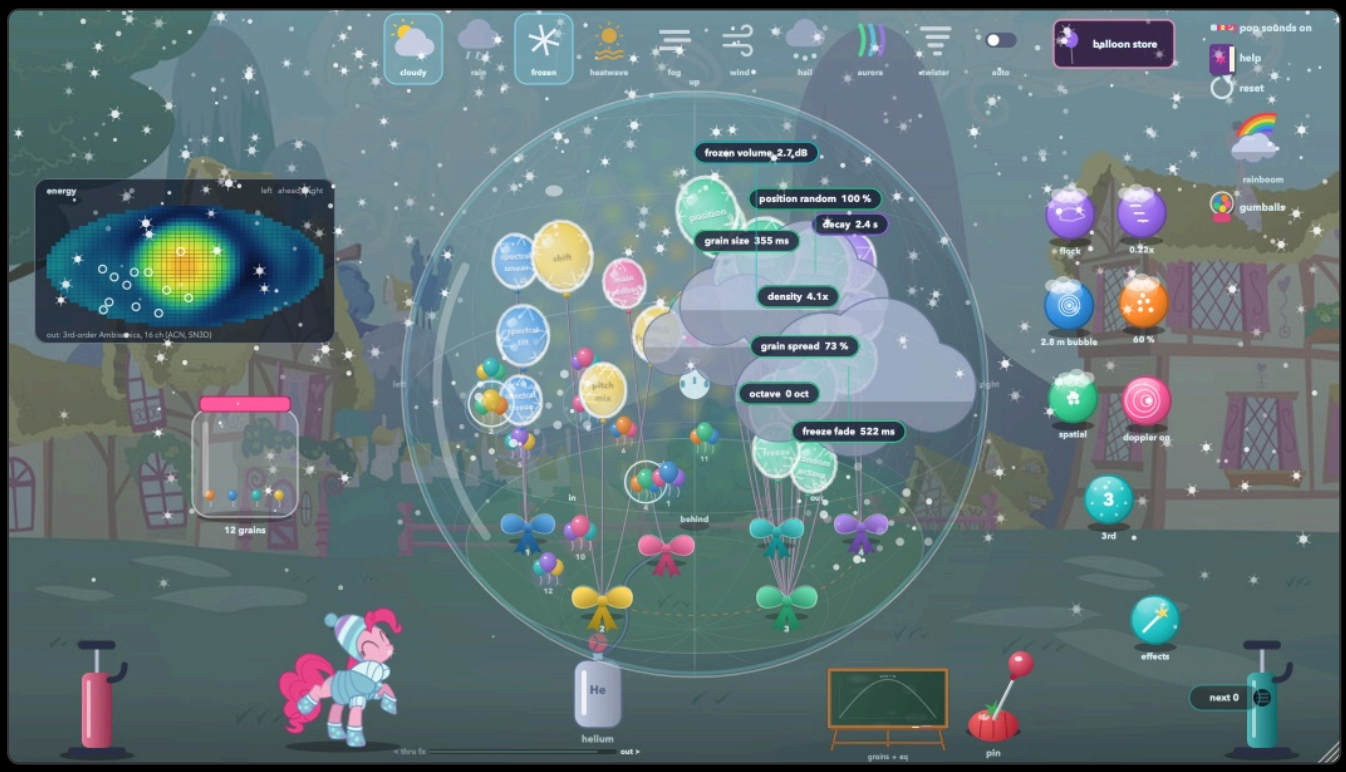
The doppler token switches that pitch bend on and off. Off, moving sounds keep their pitch and still get louder and quieter as they come and go.

When it's cloudy the tokens on the right wander too, like a balloon inside a cloud: a little cloud sits on each one while it does. Touch one and it stops.

You can drag a grain while it flies. Let go and it carries on from there.



A balloon flying into the sphere after it was let go.



A flock, with weather drifting through the glass.



Soundcube, after Bernhard Leitner: the grains step from loudspeaker to loudspeaker round the walls of a cube.



Pendulum, after Leitner: each grain swings through underneath you.



Sonotron, after Iannis Xenakis: one ring, turning faster and faster as it widens.



Nomos, after Xenakis: the grains on the corners of a turning cube.



Paraboloid, after Xenakis: straight glides that together draw a saddle.

Out

Out comes Ambisonics in ACN order with SN3D normalisation, the format the IEM and SPARTA plug-ins use. The order token (bottom right) sets how sharp it is, from 1st to 7th order: 3rd order (the default) is 16 channels, 5th is 36 and 7th is 64. A higher order pins every sound down more sharply, mostly heard on a big dome of loudspeakers. In REAPER, give the track that many channels and put a decoder after it at the same order, such as the IEM AllRADecoder for loudspeakers or the BinauralDecoder for headphones. On a track with too few channels for the order you picked, the highest order that fits comes out, and the map says so.

There is one reverb for the whole sphere (the lilac balloons), built like the IEM FdnReverb: 64 delay lines run on the 16 channels, with the highs dying away faster than the lows. The grains further out send more into it.



One grain: the energy map on the glass and at the top left shows where it is.

Colours

Each bunch has its own colour, and so does its bow.



The library's colour page: which balloon colour belongs to which effect.

The effects

Your sound runs through four effects in a row.

First the spectral smear (the blue balloons), which blurs the sound across time. Spectral freeze holds it still. Spectral tilt leans the smear from dark to bright: up lifts the highs and sinks the lows, down does the opposite, and the middle is flat. The smear stays about as loud either way.

Then the pitch shifter. Shift picks the interval, snapped to the scale chosen on the card at the top left, and pitch feedback sends it round again so it keeps climbing.

Then the grains, and a reverb that keeps the low end clean (in the sphere there's one reverb for all the grains together, at the end). That's the order they start in: the order chapter says how to change it.

Each effect has a mix balloon. When a mix balloon is empty, that effect is out of the way.

Main feedback takes the end of the chain back to the start, so the whole shop can ring on by itself. It is tamed so it sings without blowing up.

Grains

The mint bunch chops your sound into small grains. Grain size sets how long each one is and density sets how many overlap.

Shape is the envelope of each grain. The last shape, chalk, is one you draw yourself on the chalkboard.

Random pan scatters the grains across the stereo field and reverse flips some of them backwards. Octave moves them up or down, or let random octave choose for each grain.

Position picks where in the last four seconds the grains read from, from right now to four seconds back, and position random lets them wander around that spot. Click freeze to hold the sound in place. Freeze fade sets how slowly it slides in and out of the freeze.

These balloons also shape every frozen balloon you catch with the sound pump, even when the effects are bypassed.

The gumball machine

The gumball machine is sold at the balloon store. Buy it and it comes out in an orange gift box with four balloons, and takes its place on the shelf like the other effects; give it back (or pop one of its balloons) and the sound goes straight past it again.

The orange bunch is a stutter that plays with the whole sound in time with your track (or at its own steady tempo when the track is stopped). On each sixteenth note it may grab the moment and do something with it.

Stutter sets how often that happens. Bounce leans it from plain rolls and flams towards bouncing-ball repeats that speed up until they become a pitch, slowing ones that do the opposite, and frozen buzzing grains. Cut-up lets it pull its sound from other steps of the last bar, so the bar gets rearranged like a breakbeat. Sway holds back every second sixteenth, from a gentle swing to a stumble.

These ideas come from Alexander Mesker's study of Autechre's music: the bouncing ball, the event types, restarting patterns and the swing table.

Now and then a pluck from the pop pump bounces too.



The gumball machine, bought and on the shelf.

The order

The sound goes through the effects from left to right, from in to out: the spectral smear (blue), pitch (yellow), grains (mint), reverb (lilac) and a gift box for each effect from the store, the gumball machine (orange) among them.

The order list waits just off the left edge of the window. Move the cursor to the left edge and it slides out, with the effects from top (in) to bottom (out). Drag one up or down to move it: the others make room, and when you let go the bows hop to their new places and the sound follows with a tiny dip so it never clicks. The die at the top rolls a new order. The order is saved with your session.

The loop (main feedback) and the caught sounds stay on the floor: the loop wraps round all the effects.



The order list, pulled out from the left edge.



The shelf after a new order.

The pop pump

Hold the pump on the left to blow up a balloon.

While it fills, short plucked saws play on random notes of the scale, slowly and unevenly. Every pluck is folded and panned at random, and its attack is sometimes snappy and sometimes slow. The plucks go through the effects.

Somewhere between 5 and 12 seconds the balloon pops, and every pop is different.

The pop sounds switch at the top right, above help and reset, turns the hiss and the pops on or off. Right-click it to feed them into the effects instead of straight out, so the pops go through the chain with your sound.



Blowing up a plucked note with the pink pump.

Catching sounds

Hold the pump on the right to catch a sound in a balloon. It catches what comes out of the effects, or your plain input if you click the catch token beside it. The balloon starts muffled and opens up as it fills. Let go and it floats away, playing what it caught as frozen grains.

You can keep six at once, and the teal frozen volume balloon sets how loud they all are. The badge by the pump shows the pitch of the next one. Click it, or click any caught balloon, to type an interval like +7 or -5, then press return. Scrolling over a caught balloon retunes it along the scale.

To pop one, double-click it, or drag the pin from the tomato cushion onto it.



Catching a sound with the teal pump.

The canister

Click the helium canister to fill the room with helium. It adds a gentle highpass to the output, and the balloons stretch and float higher.

Slide the canister along its rail to choose where the frozen balloons go. To the right they come straight out. To the left they go back through the effects, so the pitch shifter and the reverb play with them too.



The canister slid towards the effects: the frozen balloons go round again.

Weather

The weather switches are one row along the top of the window, all the same size: cloudy, rain, snow, heatwave, fog, wind, hail, aurora, twister and auto weather. Click one to turn it on and again to turn it off. One that is on is lit up. Rain and clouds together make a storm.

Rain eases a gentle lowpass onto the output and soaks the balloons.

In the sphere the weather also drifts through the inside of the glass as patches: clouds, rain clouds, snow flurries and heat haze. Any grain a patch passes over gets its settings shaken up at random while it's in there. Snow freezes that grain's spectrum, and the heat smears it.

When it's cloudy, clouds drift through the bouquet, and any balloon inside one slowly wanders to new values, with a little tag showing where it has got to. Clouds bring thunder too, which pulls the rain's lowpass down in slow rolls.

In a storm, lightning strikes now and then and drops a random effect from the store into the room for a minute or so. If you like it, tick it in the store and it stays.

Snow freezes everything over the fade-in time: the sound slows until it stops completely and holds, and the balloons ice over and hang still. It stays frozen until you click the snowflake again, then thaws over the fade-out time. Right-click the snowflake to set both times, anywhere from 5 seconds to 5 minutes: drag along a bar or scroll on it, and double-click puts it back to 5 seconds. Leave Pinkie out in the snow for a minute and she catches a cold; she perks up once the snow has gone.

Snow and a heatwave never come together: turning one on sends the other away.

A heatwave melts the sound into a slow smear over its fade-in time (right-click the hot sun to set its fade in and fade out, like the snow's): whatever you play is spread across many seconds, and the held sound keeps drifting towards whatever comes next, like a stretched recording that never quite stops. The top end softens and swells as the hot air wobbles, gently, nothing like the rain. Pinkie can't take it and flops down flat, and if it rains in the heat she's soaked through.

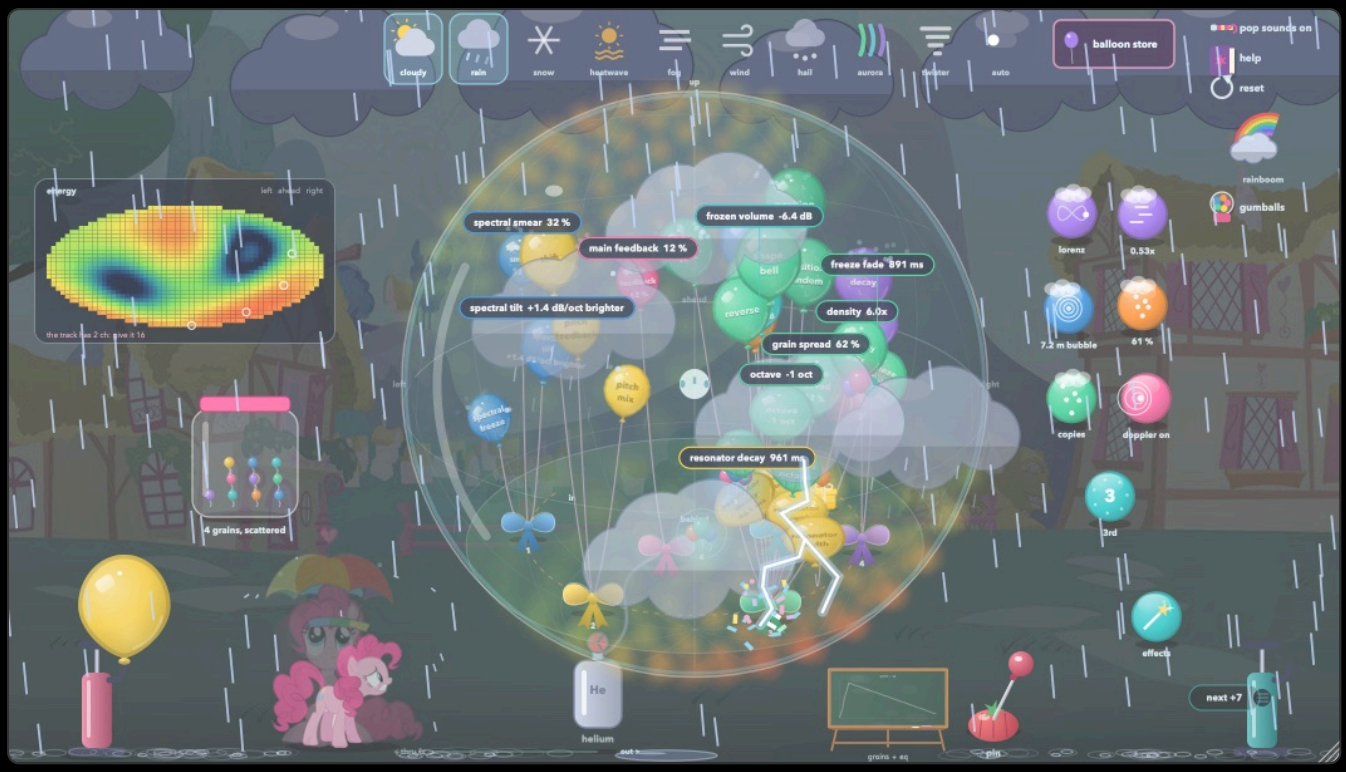
With auto weather on, the sky looks after itself. Every few minutes the weather turns, the way it does outside: clear skies cloud over, the cloud brings rain or a storm or snow, a long hot spell might end in a summer storm. Each spell lasts somewhere between a minute and twelve, and the name under the auto switch shows what the sky is doing. You can still flip the switches yourself; auto carries on from there.



Cloudy: clouds drift through the bouquet and the balloons inside them wander.



A storm: rain and clouds together.



Lightning lends an effect from the store.



Snow: everything frozen.



A heatwave.



Right-click the snow or the heat for its fade times.



Rain in the heat: Pinkie soaked through.

More weather

Fog, wind, hail, aurora and twister come after the heatwave in the weather row.

Fog blurs every attack. The sound goes through six short echoes too close together to hear apart, so a hit loses its edge without growing a tail, and the highs sink. In the sphere the fog also hides where things are. The whole scene goes a little soft at the edges, and fog banks drift through the ball: a grain inside one comes from all round you instead of from one point, duller and with more of the room on it, until it flies out again.

Wind comes in gusts. The sound is read through a delay that keeps drifting, so its pitch wavers a few cents sharp and flat, up to about a fifth of a semitone in a strong gust. The wind also blows every grain the same way across the sphere, harder in each gust, and the balloons lean over with it. The wind blows the fog away: turning one on sends the other off.

Hail throws hailstones. Each one is a tiny slice of the last moment of your sound, hit hard and then bouncing, quicker and quieter, like a ball dropped on a table. Every stone lands high overhead at its own bearing and bounces its way down to the floor.

The aurora plays your sound an octave up into a long reverb. It hangs in four curtains high up round you. They drift slowly round the sphere, and each one swells and fades by itself.

Twister starts a whirlwind that lasts between 10 and 20 seconds. It takes every grain up into one spiral, a funnel that is narrow at the floor and wide at the top, and the pitch climbs a fifth and falls back while it turns. When it is over the grains fly back to what they were doing. With wind and clouds both on, one comes by itself every few minutes.

The rainbow has no switch. When the rain stops under a clear sky (no clouds, no snow) after at least 20 seconds of rain, a rainbow stands in the sphere for about 25 seconds: seven arcs across one side of the sky, red the lowest sound and the outermost arc, violet the highest and the innermost, each colour arriving a little after the one before.

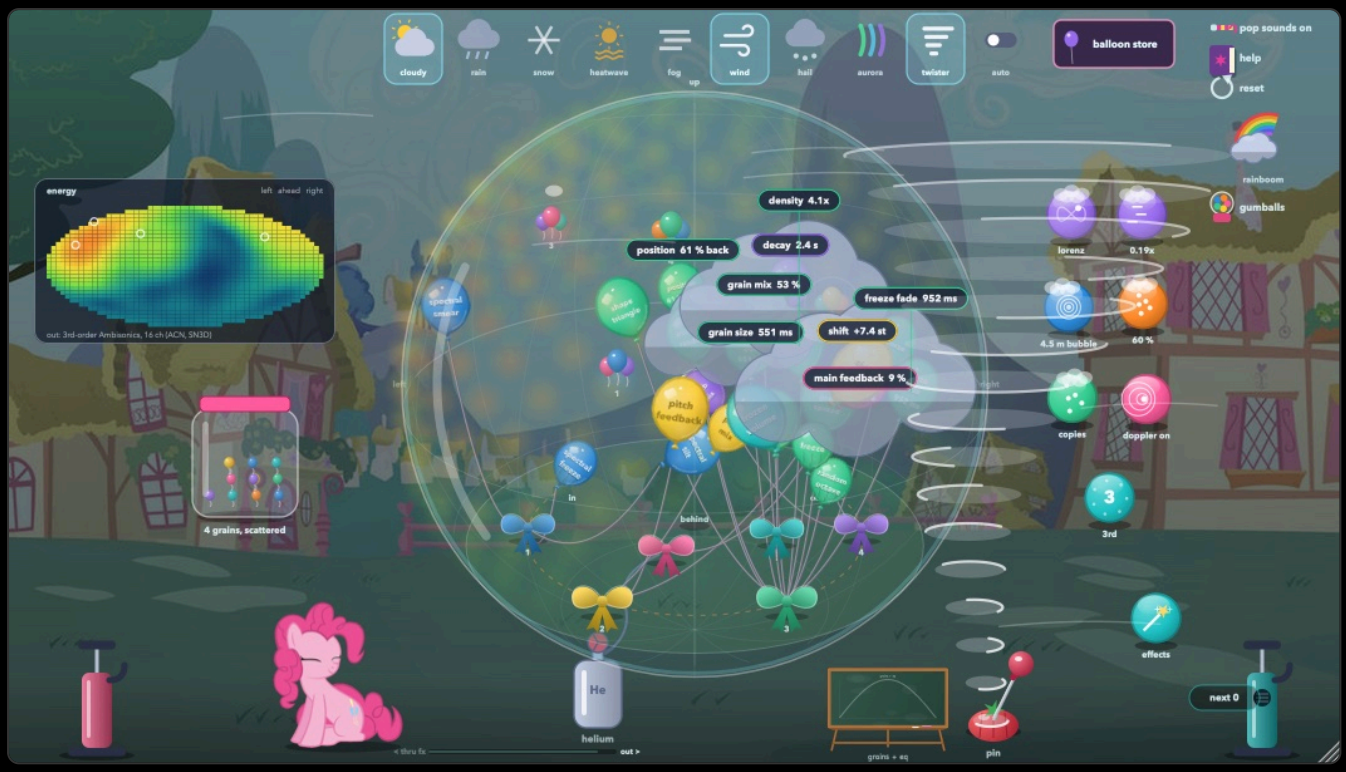
Auto weather knows all of these. Fog comes after cloud, rain or snow. Wind can blow up a storm. Hail falls out of a storm, and an aurora follows a clear spell.



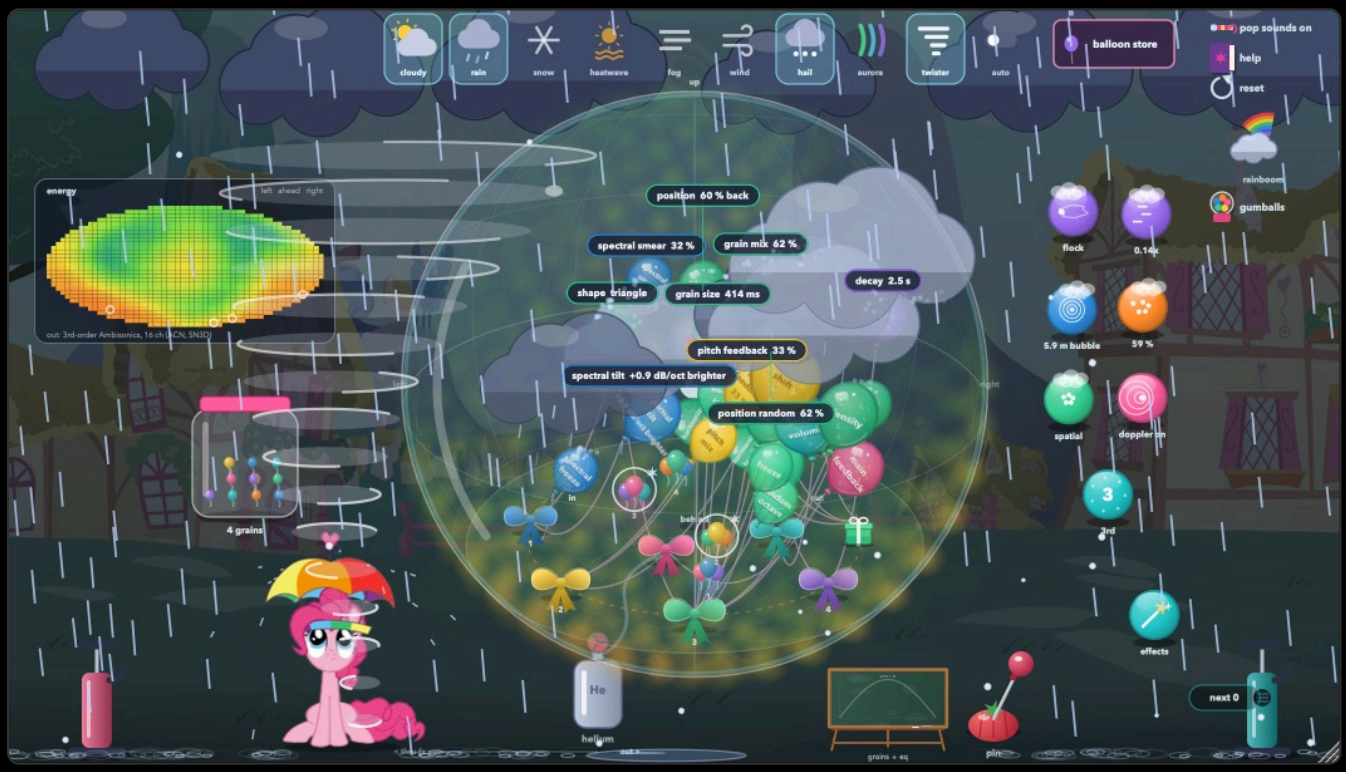
Fog.



Wind: streaks of air, and the balloons leaning over.



A twister.



Hail in a storm.



A rainbow after the rain, and the aurora across the top of the sky.

Sonic rainboom

The little cloud with a rainbow, under help and reset, is the sonic rainboom. Click it and Rainbow Dash dives in from the sky, faster and faster, the rainbow streaming out behind her, until she breaks the sound barrier: a boom and a rainbow at the same moment.

On the way in your sound is squeezed up in pitch, the way a sound rushing towards you would be, with a rush of air under it. The boom has the real shape of a sonic boom, a double crack. Then the rainbow: your sound split into seven colours, red the lowest and violet the highest. They arrive one after another, so they sweep up from the lows to the highs, and each one keeps ringing for the rainbow's length while the rainbow fades from the sky.

The shockwave knocks the balloons back, and the ones nearest it are nudged to new values. Pinkie jumps, amazed.

Fling a balloon hard enough and it breaks the barrier itself: a white cone grows behind it as you get close. Right-click the cloud for what else sets one off, such as a hit in your sound or the rain stopping after a storm. The same menu sets how loud it is and how long the rainbow lasts.

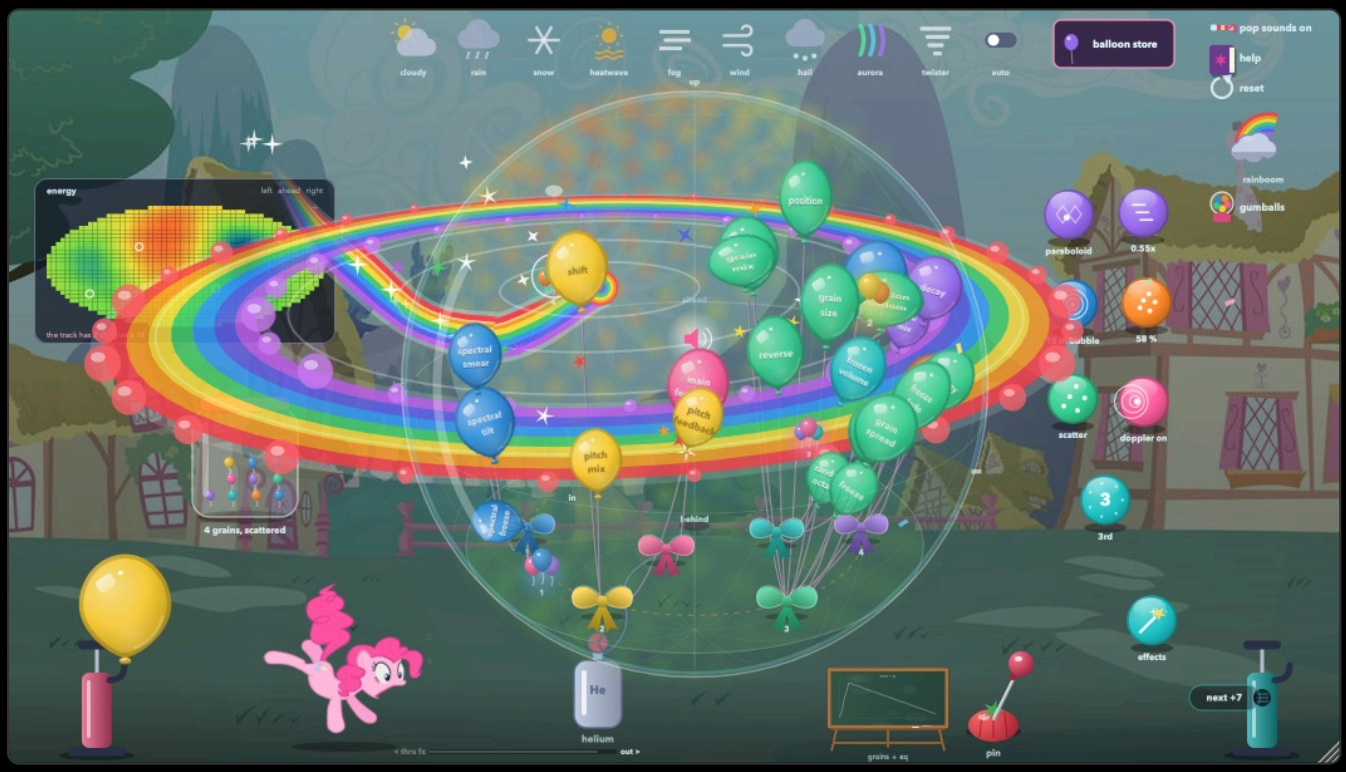
In the sphere it all happens in 3-D. The dive flies in from outside the glass to where it breaks, and the boom comes from there. Then seven rings open out flat round it, red outermost. A ring off to one side sounds like an arc over there; as it passes over you it surrounds you, and then it leaves through the glass. With on fast grains ticked, a grain flying faster than 30 metres a second breaks it where it is.



The dive: Rainbow Dash speeding in, the rainbow streaming out behind her.



The boom: the starburst and the ring of seven colours. Pinkie jumps, amazed.



The ring opening out; the shockwave has knocked the balloons back.



The rainbow left in the sky, fading over the rainbow's length.

The balloon store

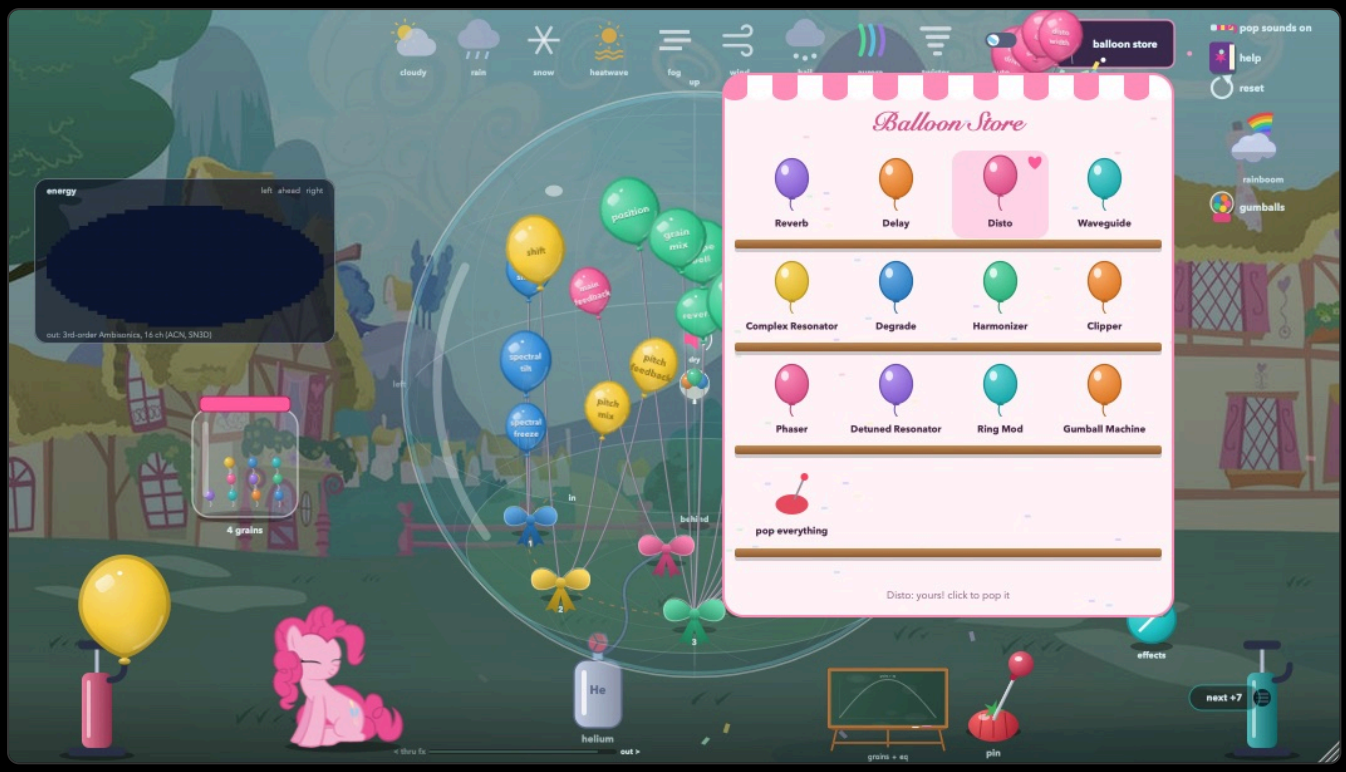
Click the balloon store sign at the top to open the store. Pick an effect and a little gift box drops out of the shop with four balloons tied to it: its two controls, then its level and its width. The box takes a place among the other effects, and the order list moves it about like the rest. Width spreads the effect from mono in the middle to wide stereo.

Pick a bought effect again to give it back, or pop one of its balloons with the pin. Pop everything sends them all back at once.

The tuned effects in the store follow the scale as well. The ring mod multiplies the sound by a tone, and with follow turned up the tone rises as the sound gets louder.



The balloon store.


cloudy
rain
snow
heatwave
fog
balloon store
pop sounds on
help
reset
rainboom
gumballs
energy
left ahead right
out: 3rd-order Ambisonics, 16-ch (ACN, SN3D)
Balloon Store
Reverb
Delay
Disto
Waveguide
Complex Resonator
Degrade
Harmonizer
Clipper
Phaser
Detuned Resonator
Ring Mod
Gumball Machine
pop everything
Disto: yours! click to pop it
4 grains
position
grain mix
shift
spectral tilt
spectral freeze
pitch feedback
pitch mix
in
behind
1
2
3
He
helium
effects
next +7
pin
grains + eq
out ▸

Its shelves.

The gumball menu

Click gumballs, under the rainboom's cloud at the top right, to open the gumball menu. It shows every control in the palace again as a gumball that stays where it is, for when you'd rather not chase a balloon. There is one row for each effect, in its balloons' colour. The sphere's own tokens (flight, speed, distance, deviate, grain mode, doppler, order and how many grains) have a row too.

Each gumball is a glass ball that fills with colour as its value goes up, with the value written inside. Drag one up or down to turn it (hold shift for fine changes), or scroll over it. Click a switch to flip it. Click a setting like shape to step to the next one, or right-click for the one before. Double-click puts a gumball back how it started. Point at one to read its full name and value.

The gumballs and the balloons are the same controls. Turn one and the other follows, and the balloons still work just as they did.

When an effect's balloons come into the room, bought at the store or dropped by the lightning, its row of gumballs pops in here. When they pop, the row pops out. With many effects bought, short rows share a line and the page scrolls.

Click the cross or press escape to go back.



The gumball menu: a row for each effect.



With the gumball machine and a store effect bought: their rows have popped in.

The chalkboard

Click the little easel on the floor to bring the chalkboard forward. It has two pages, written along the top.

On grain shape, draw across the board to make the life of one grain, from start on the left to end on the right, with loud at the top. Drawing switches the grains to your shape.

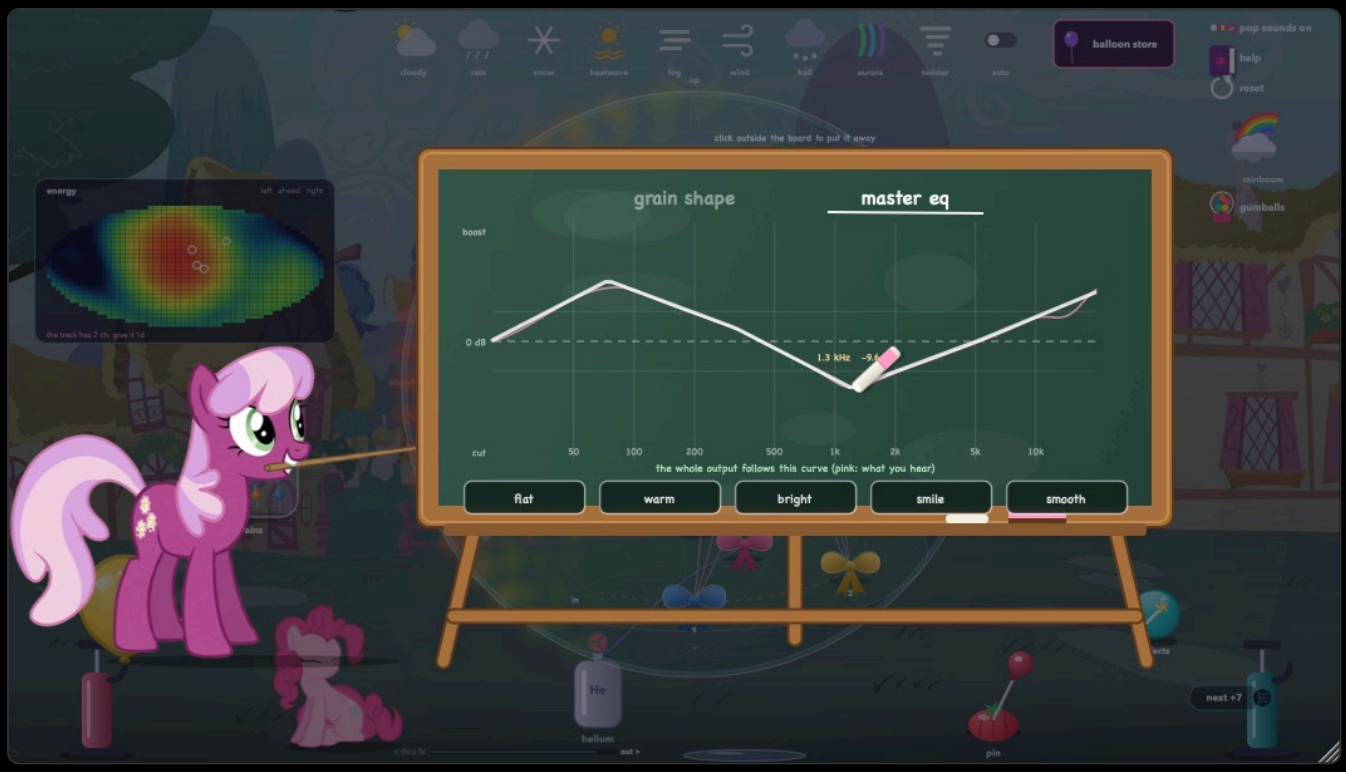
On master eq, draw the tone of the whole output. Above the dashed line boosts and below it cuts, with the lows on the left. The pink line shows what you actually hear. The buttons underneath give you ready-made curves, smooth what you drew, or set it flat again.

On master eq the board goes from a deep cut at the bottom to a big boost at the top, 24 dB each way.

Click anywhere outside the board to put it away.



Drawing a grain shape on the chalkboard.



Drawing the master EQ.

The card

The card holds the presets and the scale. It waits just above the top left of the window: move the cursor up there and it slides down. Every preset changes the picture behind the shop as well as the sound.

The scale decides which notes the pitch shifter lands on. Everything tuned in the shop follows it.

The reset button under the help book in the top right corner brings everything back to normal: the first preset, a flat EQ, the plain grain shape and the effects in their first order. Caught sounds stay where they are.

The switch underneath the card bypasses the effects, the store's ones included. Only your input and the frozen balloons are left, and the grain balloons still shape them.



The card, down from the top left corner.


cloudy
rain
snow
heatwave
fog
wind
hail
aurora
twister
auto
balloon store
pop sounds on
help
reset
rainboom
gumballs
energy
left ahead right
orbit
0.26x
7.1 m bubble
35 %
spatial
doppler on
3rd
4 grains
dry
out
in
behind
start
feedback
stretch
useful
scroll or drag along ↔
Sugarcube Corner
Cloudsdale Carousel
Butterfly Falls
Flock Over Ponyville
Canterlot Chimes
Moonlight Howl
Smokey Mountain Squall
Las Pegasus Neon Loop
Cloudsdale Stretch

The presets.

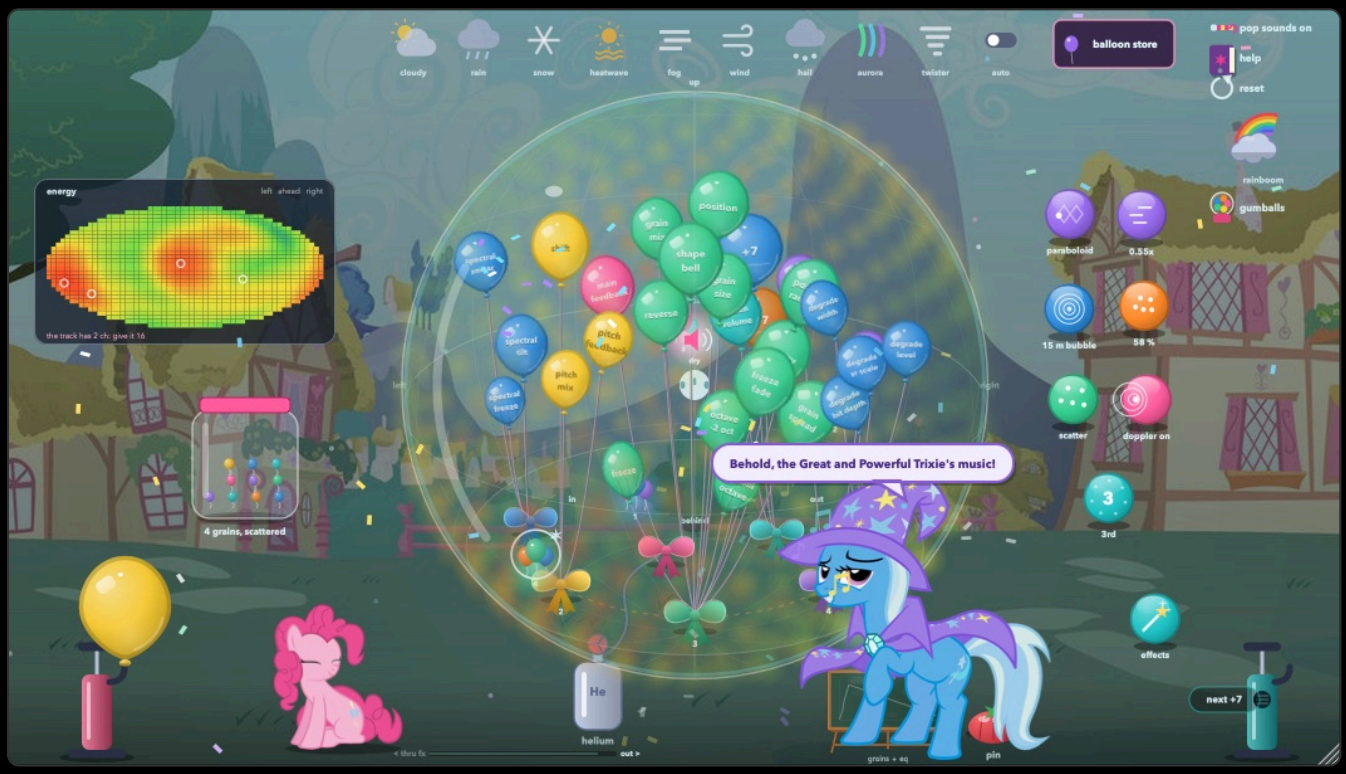
Pinkie

Pinkie sits by the pop pump watching the balloons.

Pops make her jump. When it rains she gets her umbrella out (unless it's a heatwave too: then she lies in the puddle, soaked), and helium turns her to crystal.

Leave the shop alone for a while and she falls asleep.

Every so often, about once an hour, the Great and Powerful Trixie barges in. She zaps balloons from her amulet, each one jumping to a new value, then sets everything she can reach at random with one big flash and leaves. Some visits she also plays a strange synthesised noise into the shop's input, which goes through all the effects, with a chirp for every zap. Sometimes she zaps a few effects from the store into the room too, and those stay. She leaves the bypass and the volume alone, and never touches the snow, the heat or auto weather. If you don't like what she did, the reset button under the help book puts everything back to normal.



Trixie drops by.



Leave the shop alone for a while and Pinkie falls asleep.

The flight modes

Pick one with the flight token; speed sets how fast (0.05x to 3x).

still	Each grain stays where you dropped it.
orbit	Each grain circles you on its own tilted ring.
lorenz	Along the Lorenz attractor (the butterfly), each grain starting somewhere else on it.
flock	A flock of birds (boids): they keep apart, line up and stay together.
random	A new place every so often, gliding there.
spiral	Winding round you, climbing and sinking.
lissajous	Looping figures, each grain at its own rates.
rossler	The Rossler attractor: a loose coil that now and then flicks up.
breathe	Out to the glass and back in, like breathing.
fall	From the top of the sphere to the bottom, over and over.
ping-pong	Straight across the sphere, from one side to the other.

After Bernhard Leitner

Since 1968 Leitner has built rooms out of sound moving along lines of loudspeakers: sound as a building material, drawn through a space and through the body. These four flights take up his figures.

line	A line of sound through the room and through you: each grain runs along its own straight line from wall to wall and back, at an even pace.
pendulum	A swinging space: each grain hangs from a point overhead and swings through underneath you, slow at the ends and quick in the middle.
column	A vertical space: each grain climbs and sinks along its own column standing round you, winding a little round it.
soundcube	A cube with nine loudspeakers on each wall. The grains don't fly: they step from one loudspeaker to the next, drawing a ring round you, then another higher or lower. Every third grain draws its rings round a different axis.

After Iannis Xenakis

Xenakis moved sound through space with the same mathematics he composed with. These five flights each take one of his procedures and apply it to where the grains are.

sonotron	All the grains on one ring that turns faster and faster along a logarithmic spiral as it widens, then starts again the other way (Terretektorh, the end of Persephassa).

paraboloid	Every grain glides along its own straight line; the lines are the two families of rulings of a hyperbolic paraboloid, z = xy (the Philips Pavilion, the glissandi of Metastaseis).
gas	The grains are the molecules of a gas: each flies straight, its speed drawn from the Maxwell-Boltzmann distribution, and leaves the glass at a new speed when it hits it (Pithoprakta). Speed is the temperature.
nomos	The grains sit on the eight corners of a cube (a second eight on a smaller cube inside). The cube turns from one of its 24 rotations to the next, each the product of the two before it (Nomos Alpha).
screens	The sphere is cut into eight cells. Every moment each grain moves on by a Markov chain, one two-state chain per axis, with the transition tables of Analogique A (0.2 / 0.8 and 0.85 / 0.4); which table an axis uses depends on where the grain is on the other two.

The wind pushes the grains of any flight the same way while it blows, and a twister takes them all up into its spiral until it is over.

Distance and Doppler

The distance token is the size of the bubble you're in: its radius, from 1 metre to 50. Every sound is a point with its own distance from you, and distance sounds the way it does in a room, after the IEM RoomEncoder:

- Level falls with distance (1 / r), referred to 4 metres, so a sound 4 m away is at its own level and nearer ones get up to 12 dB louder. A 1 m bubble is about 24 dB louder than a 50 m one.
- Sound takes time to arrive, at 343 metres a second, so a sound 50 m away arrives 146 ms late. Anything moving towards you or away bends in pitch, like a passing car (Doppler). The Doppler token switches that bend off: moving sounds then keep their pitch and still get louder and quieter as they come and go.
- Air takes a little of the highs off with distance, and six walls of a room round the bubble answer each sound faintly (9 dB down), enough to tell how far without blurring where.
- Close in, a sound spreads round your head instead of pointing.

The bubble glides when you turn it, so with Doppler on, resizing it bends the pitch for a moment, as it would.

Grain modes

spatial	The default. Every grain carries its own copy of the effects, each copy's sound a point at its grain, its settings deviating from yours. Your dry sound plays once, from the loudspeaker.
scatter	One copy of the effects, at the first grain; the granulator's grains still fly to every grain.
copies	Every grain is the whole shop, dry sound and all.

With more grains each is turned down (by 1 / sqrt(N)), so the whole stays about as loud. 16 grains in spatial mode take about 20 to 24 % of one core at 48 kHz on an Apple M2 Max.

Quick reference: what your hands do

Drag a balloon	Move it. Let go and it floats back up on its string.
Scroll or swipe on a balloon	Blow it up or let air out (its value).
Shift-drag up or down	Set it by hand. Add alt for finer steps.
Double-click a balloon	Back to its default.
Click a switch or a choice	Switches flip; choices like shape step to the next.
Drag a bow	Move its whole bunch. Along the shelf, it changes the order.
Drag the pin onto a balloon	Pop it (caught sounds, store effects).
Click a caught sound	Type its pitch.
Right-click the snow or the heatwave	Their fade times.
Right-click pop sounds	Pops on or off, or into the effects.
Click the rainboom cloud	A rainboom now. Right-click: what sets them off, level and the rainbow's length.
Fling a balloon hard	Breaks the sound barrier: a rainboom where it's got to (when on throws is ticked).
Click gumballs (top right)	The gumball menu: every control as a gumball that stays put. Drag or scroll one to turn it, click a switch, double-click to reset. Escape closes it.
Click twister	A whirlwind now, for 10 to 20 seconds.
Cursor to the top left corner	The card: presets, scale and bypass.
Cursor to the left edge	The order list. Drag an effect up or down, or roll the die.
Drag the empty glass	Turn the sphere (your view only, never the sound).
Scroll or pinch on the glass	Zoom in towards the cursor.
Double-click the glass	Reset the view.
Click the jar	Another grain. Drag one out to drop it somewhere; drag it back or right-click the jar to take one away.
Drag the pink loudspeaker	Move your dry sound in the sphere.
Drag or scroll a token	Turn it like a knob; click a choice token to step it.

Parameter reference

Every control your DAW can automate, by the name it shows there. Defaults are what a fresh instance starts with (reset brings them back). The store's effects follow on the next pages.

The sphere

The tokens on the right and the jar.

Name in your DAW	ID	Range	Default
Grain Mode	`grainMode`	spatial, scatter, copies	spatial
Grains	`grains`	1 grain to 16 grains	4 grains
Deviate	`deviate`	0 % to 100 %	35 %
Flight	`flight`	still, orbit, lorenz, flock, random, spiral, lissajous, rossler, breathe, fall, ping-pong, line, pendulum, column, soundcube, sonotron, paraboloid, gas, nomos, screens	orbit
Flight Speed	`flightSpeed`	0.05x to 3.0x	0.26x
Distance	`spread`	1.0 m bubble to 50 m bubble	7.1 m bubble
Doppler	`doppler`	off / on	On

Spectral smear (blue)

Name in your DAW	ID	Range	Default
Spectral Mix	`sgMix`	0 % to 100 %	35 %
Spectral Freeze	`sgFreeze`	off / on	Off
Spectral Tilt	`sgTilt`	-6.0 dB/oct darker to +6.0 dB/oct brighter	flat

Pitch (yellow)

The scale is on the card.

Name in your DAW	ID	Range	Default
Pitch Shift	`psShift`	-24.0 st to +24.0 st	+7.0 st
Pitch Feedback	`psFeedback`	0 % to 95 %	35 %
Pitch Mix	`psMix`	0 % to 100 %	30 %
Scale	`psScale`	Major (Ionian), Natural Minor (Aeolian), Harmonic Minor, Melodic Minor, Dorian, Phrygian, Lydian, Mixolydian, Locrian, Major Pentatonic, Minor Pentatonic, Blues, Phrygian Dominant, Whole Tone, Diminished (Whole-Half), Chromatic, The Well-Tuned Piano (La Monte Young), Raga Bhairav, Raga Yaman, Raga Bhairavi, Raga Todi, Raga Marwa, Raga Purvi, Raga Kafi, Raga Malkauns, Raga Darbari Kanada, Raga Bhupali, Raga Desh, Raga Puriya Dhanashri, 22 Shruti, Maqam Rast, Maqam Bayati, Maqam Saba, Slendro, Pelog, Harmonic Series 8-16, Partch 43-Tone, Bohlen-Pierce (tritave), Quarter-tone (24-EDO), 19-EDO, 31-EDO	Major (Ionian)

Grains (mint)

These also shape the frozen sound balloons.

Name in your DAW	ID	Range	Default

Grain Mix	`grMix`	0 % to 100 %	40 %
Grain Size	`grSize`	10 ms to 1.50 s	120 ms
Grain Density	`grDensity`	1.0x to 24.0x	7.0x
Grain Shape	`grShape`	saw, reverse saw, bell, triangle, pulse, sine, chalk	bell
Grain Spread	`grPan`	0 % to 100 %	25 %
Grain Reverse	`grReverse`	0 % to 100 %	0 %
Grain Position	`grPos`	0 % back to 100 % back	50 % back
Grain Position Random	`grPosRand`	0 % to 100 %	100 %
Grain Octave	`grOctave`	-2 oct to +2 oct	0 oct
Random Octave	`grRandOct`	off / on	On
Freeze	`grFreeze`	off / on	Off
Freeze Fade	`grFade`	10 ms to 10.00 s	1.00 s

Reverb (lilac) and the loop (pink)

Name in your DAW	ID	Range	Default
Reverb Decay	`rvDecay`	0.2 s to 20.0 s	4.0 s
Reverb Mix	`rvMix`	0 % to 100 %	30 %
Main Feedback	`masterFb`	0 % to 100 %	0 %

The gumball machine (orange)

Bought at the balloon store.

Name in your DAW	ID	Range	Default
Stutter Bought	`stOwned`	off / on	Off
Stutter	`stAmount`	0 % to 100 %	0 %
Stutter Bounce	`stBounce`	0 % to 100 %	50 %
Stutter Sway	`stSway`	0 % to 100 %	0 %
Stutter Cut-up	`stCutup`	0 % to 100 %	25 %

Sound balloons, pops and the canister

Name in your DAW	ID	Range	Default
Catch From	`catchFrom`	effects, input	effects
Frozen Balloons Volume	`grainsLevel`	silent to 6.0 dB	0.0 dB
Frozen Grains Through FX	`grainsThru`	straight out to through fx	straight out
Balloon Sounds	`balloonSfx`	off / on	On
Pop Sounds Through FX	`popsThru`	off / on	Off

Weather

Name in your DAW	ID	Range	Default
Cloudy	`cloudy`	off / on	Off
Rain	`rain`	off / on	Off

Snow	`snow`	off / on	Off
Heatwave	`heat`	off / on	Off
Helium	`helium`	off / on	Off
Auto Weather	`autoWeather`	off / on	Off
Snow Fade In	`snowIn`	5 s to 5 min	5 s
Snow Fade Out	`snowOut`	5 s to 5 min	5 s
Heatwave Fade In	`heatIn`	5 s to 5 min	5 s
Heatwave Fade Out	`heatOut`	5 s to 5 min	5 s
Fog	`fog`	off / on	Off
Wind	`wind`	off / on	Off
Hail	`hail`	off / on	Off
Aurora	`aurora`	off / on	Off
Twister	`twister`	off / on	Off

Sonic rainboom

Name in your DAW	ID	Range	Default
Rainboom	`rbBoom`	off / on	Off
Rainboom Level	`rbLevel`	0 % to 100 %	70 %
Rainbow Length	`rbTail`	1.0 s to 12.0 s	5.0 s
Rainboom On Throws	`rbThrows`	off / on	On
Rainboom On Hits	`rbHits`	off / on	Off
Rainboom After Storms	`rbStorms`	off / on	Off
Rainboom On Fast Grains	`rbFast`	off / on	Off

Bypass

Name in your DAW	ID	Range	Default
Bypass Effects	`fxBypass`	off / on	Off

Other

Name in your DAW	ID	Range	Default
Ambisonic Order	`ambiOrder`	1st, 2nd, 3rd, 4th, 5th, 6th, 7th	3rd

The balloon store's effects

Each effect has a *Bought* switch (whether it's on the shelf), its two controls, a level (up to 200 %) and a width (from mono in the middle to wide stereo). The tuned ones follow the scale.

Reverb

Name in your DAW	ID	Range	Default
Reverb Bought	`stReverbOwned`	off / on	Off
Reverb Feedback	`stReverbA`	0.00 to 1.00	0.75
Reverb Cutoff	`stReverbB`	100 Hz to 15.00 kHz	5.00 kHz
Reverb Level	`stReverbLevel`	0 % to 200 %	100 %
Reverb Width	`stReverbWidth`	mono to stereo	50 % wide

Delay

Name in your DAW	ID	Range	Default
Delay Bought	`stDelayOwned`	off / on	Off
Delay Delay	`stDelayA`	10 ms to 1.0 s	250 ms
Delay Feedback	`stDelayB`	0.00 to 1.00	0.50
Delay Level	`stDelayLevel`	0 % to 200 %	100 %
Delay Width	`stDelayWidth`	mono to stereo	50 % wide

Disto

Name in your DAW	ID	Range	Default
Disto Bought	`stDistoOwned`	off / on	Off
Disto Drive	`stDistoA`	0.00 to 1.00	0.75
Disto Slope	`stDistoB`	0.00 to 0.99	0.75
Disto Level	`stDistoLevel`	0 % to 200 %	100 %
Disto Width	`stDistoWidth`	mono to stereo	50 % wide

Waveguide

Name in your DAW	ID	Range	Default
Waveguide Bought	`stWaveguideOwned`	off / on	Off
Waveguide Frequency	`stWaveguideA`	20.6 Hz to 494 Hz	98.0 Hz
Waveguide Fall time	`stWaveguideB`	1.0 s to 60.0 s	30.0 s
Waveguide Level	`stWaveguideLevel`	0 % to 200 %	100 %
Waveguide Width	`stWaveguideWidth`	mono to stereo	50 % wide

Complex Resonator

Name in your DAW	ID	Range	Default
Complex Resonator Bought	`stComplexResOwned`	off / on	Off
Complex Resonator Frequency	`stComplexResA`	20.6 Hz to 3.95 kHz	494 Hz

Complex Resonator Decay	stComplexResB	1 ms to 5.0 s	1.0 s
Complex Resonator Level	stComplexResLevel	0 % to 200 %	100 %
Complex Resonator Width	stComplexResWidth	mono to stereo	50 % wide

Degrade

Name in your DAW	ID	Range	Default
Degrade Bought	stDegradeOwned	off / on	Off
Degrade Bit depth	stDegradeA	2.0 bits to 32.0 bits	8.0 bits
Degrade Sr scale	stDegradeB	0.01 to 1.00	0.25
Degrade Level	stDegradeLevel	0 % to 200 %	100 %
Degrade Width	stDegradeWidth	mono to stereo	50 % wide

Harmonizer

Name in your DAW	ID	Range	Default
Harmonizer Bought	stHarmonizerOwned	off / on	Off
Harmonizer Transpose	stHarmonizerA	-12 st to +12 st	-7 st
Harmonizer Feedback	stHarmonizerB	0.00 to 1.00	0.25
Harmonizer Level	stHarmonizerLevel	0 % to 200 %	100 %
Harmonizer Width	stHarmonizerWidth	mono to stereo	50 % wide

Clipper

Name in your DAW	ID	Range	Default
Clipper Bought	stClipperOwned	off / on	Off
Clipper Threshold	stClipperA	0.00 to 0.25	0.10
Clipper Cutoff	stClipperB	100 Hz to 15.00 kHz	5.00 kHz
Clipper Level	stClipperLevel	0 % to 200 %	100 %
Clipper Width	stClipperWidth	mono to stereo	50 % wide

Phaser

Name in your DAW	ID	Range	Default
Phaser Bought	stPhaserOwned	off / on	Off
Phaser Lfo freq	stPhaserA	0.05 Hz to 10 Hz	0.30 Hz
Phaser Feedback	stPhaserB	0.00 to 1.00	0.50
Phaser Level	stPhaserLevel	0 % to 200 %	100 %
Phaser Width	stPhaserWidth	mono to stereo	50 % wide

Detuned Resonator

Name in your DAW	ID	Range	Default
Detuned Resonator Bought	stAllpassWGOowned	off / on	Off
Detuned Resonator Frequency	stAllpassWGA	20.6 Hz to 494 Hz	98.0 Hz
Detuned Resonator Detune	stAllpassWGB	0.00 to 1.00	0.50

Detuned Resonator Level	stAllpassWGLevel	0 % to 200 %	100 %
Detuned Resonator Width	stAllpassWGWidth	mono to stereo	50 % wide

Ring Mod

Name in your DAW	ID	Range	Default
Ring Mod Bought	stRingModOwned	off / on	Off
Ring Mod Frequency	stRingModA	20.6 Hz to 5.27 kHz	440 Hz
Ring Mod Follow	stRingModB	0.00 to 1.00	0.30
Ring Mod Level	stRingModLevel	0 % to 200 %	100 %
Ring Mod Width	stRingModWidth	mono to stereo	50 % wide

Presets

Open them from the card at the top left (or your DAW's preset list). Each has its own picture behind the shop.

1. Sugarcube Corner
2. Cloudsdale Carousel
3. Butterfly Falls
4. Flock Over Ponyville
5. Canterlot Chimes
6. Moonlight Howl
7. Smokey Mountain Squall
8. Las Pegasus Neon Loop
9. Cloudsdale Stretch
10. Still Lake
11. Slow Rainbow Falls
12. Golden Hour
13. Ponyville Morning
14. Main Street Shimmer
15. Ghastly Gorge Echo
16. Froggy Bottom Chorus
17. Rock Farm Crunch
18. Rainbow Resonators

Problems and answers

I only hear one or two channels, or the map says the track has 2 channels

The track needs as many channels as the order: 16 for 3rd order, 36 for 5th, 64 for 7th. In REAPER, the track's Route button, Track channels. Then put a decoder after the Balloon Palace.

It sounds wrong through my decoder

Set the decoder to the same order as the Balloon Palace's order token, ACN channel order, SN3D normalisation (AmbiX). IEM's decoders use ACN and SN3D by default; in SPARTA pick ACN and SN3D.

macOS says it can't be opened, or the DAW skips it

The builds aren't notarised. Run the `xattr` line from Installing, then rescan. To check the Audio Unit yourself: `auval -v aufx Fkps Frkp`.

The sound changes on its own

Something in the weather is doing it. Clouds make the balloons inside them wander (and in the Balloon Palace the tokens too), auto weather changes the sky every few minutes, a storm's lightning lends a store effect for a while, and Trixie drops by now and then. Turn off cloudy and auto weather for a steady patch.

A rainbow or a twister came by itself

A rainbow follows any shower of 20 seconds or more that ends under a clear sky. A twister comes every few minutes while wind and cloudy are both on. Turn off the wind, or keep the clouds while the rain stops, to go without.

Everything has stopped or gone still

It's snowing: snow freezes the sound until you click the snowflake again. Also check the bypass on the card.

A rainboom went off by itself

Right-click the rainboom cloud: on hits, after storms, on fast grains and on throws each set one off. Untick the ones you don't want.

A new instance doesn't start from the defaults

Every new instance starts from the last patch you had open. A saved session still opens as you saved it. Reset (top right) puts everything back to the first preset.

It's loud, or a rainboom is too much

The limiter keeps the output under -1 dBFS whatever happens. Turn the rainboom down in its menu (level).

Credits and license

This is an unofficial fan project, not affiliated with or endorsed by Hasbro. My Little Pony and its characters are trademarks of Hasbro. The character pictures and backgrounds are fan-made, as is the "Magic is Friendship" title font, and they belong to their creators. The flying Rainbow Dash is by Nightmare Moon.

The balloon-store effects are rebuilt from Olivier Bélanger's SoundGrain (the phaser replaces its flanger). The gumball machine, the ring mod and the bouncing plucks take their ideas from Alexander Mesker's thesis *Analysis and recreation of key features in selected Autechre tracks from 1998-2005* (Macquarie University, 2007).

The ambisonic conventions follow the IEM Plug-in Suite and SPARTA; the energy map follows the IEM EnergyVisualizer; the distance model follows the IEM RoomEncoder and the reverb the IEM FdnReverb. The source-spread and distance ideas were checked against s3g-dsp. No code from these is included.

Four flight modes follow the sound-space work of Bernhard Leitner (lines and swinging spaces of sound, vertical spaces, the Soundcube of 1969). Five follow Iannis Xenakis: Terretektorh and Persephassa, the Philips Pavilion and Metastaseis, Pithoprakta, Nomos Alpha and Analogique A, as he describes them in *Formalized Music*. They are tributes to those ideas, not reconstructions of the works.

Built with JUCE. The code is released under the GNU Affero General Public License v3, as JUCE's free licence requires. The artwork is not covered by it.

Source, releases and issues: github.com/femifleming/PinkiePiesBalloonShop.